

Learn HTML: Beginner's Guide

Lesson 2: Essential Tags and Elements

Essential Tags and Elements – Study Notes

Key Concepts

1. **HTML Headings** (`<h1>` `<h6>`) – Define the hierarchy and structure of page titles and sections.
2. **Paragraphs** (`<p>`) – Contain blocks of text; the default container for readable content.
3. **Hyperlinks** (`<a>`) – Create clickable links to other pages, sections, or resources.
4. **Images** (`<img>`) – Embed visual content; require `src` and `alt` attributes.
5. **Lists** (`<ul>`, `<ol>`, `<li>`) – Organize items as unordered (bullets) or ordered (numbers).

Summary

HTML is built from a set of **semantic tags** that tell browsers (and developers) what each piece of content represents. Headings (`<h1>` `<h6>`) give a document a clear outline, making it easier for users and search engines to understand the page structure. Paragraphs (`<p>`) group related sentences into readable blocks, while the anchor tag (`<a>`) turns text or other elements into hyperlinks, connecting pages together. Images (`<img>`) add visual context; they are self-closing elements that must include a source (`src`) and an alternative text (`alt`) for accessibility and SEO. Finally, lists (`<ul>` for unordered, `<ol>` for ordered) and their list items (`<li>`) help present information in a logical, scannable format.

Understanding these fundamental tags enables you to construct clean, accessible, and well-structured web pages—an essential skill for any beginner web developer.

Important Points

- **Heading hierarchy matters:**
 - `<h1>` is the most important (usually the page title).
 - Use headings sequentially (`<h2>` under `<h1>`, `<h3>` under `<h2>`, etc.) for logical structure and SEO benefits.
- **Paragraphs automatically add spacing** before and after the block, making text readable without extra CSS.
- **Anchor** (`<a>`) attributes:
 - `href` – URL or fragment identifier (`#section`).
 - `target="_blank"` – Opens link in a new tab/window.
 - `rel="noopener noreferrer"` – Recommended when using `target="_blank"` for security.
- **Image** (`<img>`) requirements:
 - `src` – Path to the image file.
 - `alt` – Descriptive text for screen readers and when the image fails to load.
 - Optional: `width`, `height`, `loading="lazy"` for performance.

- **Lists:**

- `<ul>` creates bullet points; `<ol>` creates numbered items.
- Each item must be wrapped in `<li>`.
- Nested lists are allowed by placing another `<ul>`/`<ol>` inside an `<li>`.
 - All tags are **case insensitive** in HTML5, but the convention is lowercase.

Examples

Example 1 – Simple Content Block

```
```html
<h1>Welcome to My Portfolio</h1>
<p>Hello! I'm Alex, a front end developer. Below you'll find some of my favorite projects.</p>

 Project One
 Project Two
 Project Three

```

Explanation
```

- `<h1>` gives the page its main title.
- `<p>` wraps the introductory text.
- The `<ul>` creates a bulleted list of links; each `<li>` contains an `<a>` tag pointing to a project page.

Example 2 – Image with Caption and Ordered Steps

```
```html
<h2>How to Bake a Cake</h2>

 Preheat the oven to 350°F (175°C).
 Mix flour, sugar, and eggs in a bowl.
 Pour batter into a greased pan.
 Bake for 30-35 minutes.

<p>Here's a photo of the finished cake:</p>

```

Explanation
```

- `<h2>` introduces a new section under the main heading.

- `<ol>` numbers each step, making the process easy to follow.
- The `<img>` tag displays a picture; `alt` describes the image for accessibility, `width` controls display size, and `loading="lazy"` defers loading until needed.

Quick Review

1. **What is the purpose of using heading tags in order (`<h1>` !' `<h6>`)? **
2. **Which attribute is mandatory for an `<img>` tag and why? **
3. **How do you make a link open in a new browser tab safely? **
4. **When would you choose `<ul>` over `<ol>` and vice versa? **
5. **Write a short snippet that creates a numbered list of three favorite books, each title linking to its Amazon page. **

Further Reading

- **Semantic HTML:** `<section>`, `<article>`, `<nav>`, `<header>`, `<footer>`
- **HTML Accessibility:** ARIA roles, `role` attribute, keyboard navigation
- **Responsive Images:** `srcset`, `<picture>` element, `sizes` attribute
- **CSS Basics:** Styling headings, paragraphs, links, and lists
- **HTML Forms:** `<form>`, `<input>`, `<label>`, `<button>`

Happy coding! Ø=Þ€

